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Applied AI, Cognitive Science & Human Cognition | MScEng Human-Centered Artificial Intelligence | DTU

PROJECTS

Personalized Multi-Modal Agent System for Bias Analysis – Agentic LLMs & MM-RAG

INFS4205 Advanced Techniques for High Dimensional Data, University of Queensland

- Built a **multi-modal RAG pipeline** with a custom **web UI** over a YouTube watch-history dataset, where the original **987-row user history** was enriched with 50 custom synthetic rows and thumbnail-derived embeddings were fused with text into a shared vector representation stored in **ChromaDB** after comparing against **FAISS**
- Implemented a **LangGraph** and **LangChain** agent stack with task-specific **sub-agents** and **tools** for retrieval and bias-analysis queries, using **MM-RAG** to answer questions from the retrieved watch history and the **fused thumbnail+text multi-modal index**
- Generated benchmark questions for 4 **factual**, 4 **cross-modal**, 4 **analytical**, and 4 **conversational** items, then scored answers with an **LLM-as-Judge** setup across four ablations (vanilla, text-RAG, MM-RAG, agentic MM-RAG) with average correctness between **0.49** and **0.73**.

Link Prediction and Centrality Measure in Social Graphs – Graph ML

INFS7450 Social Media Analytics, University of Queensland

- Benchmarked several link-prediction methods on a graph with **2,485 nodes** and **4,969 edges**: **Common Neighbors**, **Spectral Clustering** with the **Normalized Laplacian**, and a 2-layer **GCN** with degree-based node features
- Constructed a **reproducible evaluation pipeline** with balanced 1:1 training/evaluation splits, and top-100 ranking over 1,000 test edges under a realistic 1:9 class imbalance
- Trained the GCN with **ReLU**, **dropout**, **binary cross-entropy**, and **Adam**, then compared methods by accuracy: **0.9949** for Common Neighbors, **0.8933** for spectral clustering, and **0.7263** for GCN; selected spectral clustering as the best trade-off for final ranking

Regression and Classification on Auto MPG – traditional ML pipeline

02450 Introduction to Machine Learning and Data Mining, DTU

- Designed a fuel consumption **regression** pipeline on a dataset with **392 observations** featuring **standardization**, **correlation analysis**, **pairplots**, **boxplots**, **histograms**, and **PCA via SVD**; then trained **ridge regression (L2)** with **10-fold cross-validation** over **50 regularization values**, and compared against an **ANN** under **10x10 nested cross-validation** using **MSE** as metric
- Carried out a 3-class origin **classification** task using **multinomial logistic regression (L2)** and an **ANN** with the number of hidden units selected by cross-validation, then evaluated with **10x10 nested cross-validation accuracy** and a **McNemar's** test between the models' performance

EEG Signal Processing for Motor Imagery Classification – BCI pipeline

22053 Principles of Brain-Computer Interfaces, DTU

- Built an EEG motor-imagery pipeline with 50Hz **notch filtering**, 12th-order **Chebyshev Type II band-pass filtering** over 8–30Hz, and **spatial filtering** using both **PCA** and **CSP** to reduce multichannel EEG into low-dimensional components
- Extracted time-frequency features with **STFT**, reduced them to band-averaged magnitudes over the **mu/beta bands**, and compared **FNN** and **CNN** classifiers under **cross-validation**; with the best pipeline reaching about **0.75** validation accuracy

Manipulation of Belief Effects on AI-Generated Content – Cognitive Science experiment

02455 Experiment in Cognitive Science, DTU

- Designed a **within-subjects experiment** with **22 participants**, using counterbalanced “AI-generated” vs “human-made” labels on the same four animal videos to isolate belief effects
- Collected **HRV (RMSSD)** and **SAM** ratings for valence, arousal, and dominance, then analyzed the data with **Shapiro Wilk**, **Wilcoxon signed-rank tests**, and 10,000-sample bootstrapped 95% confidence intervals

EDUCATION

MScEng Human-Centered Artificial Intelligence

DTU, Denmark

February 2025 – Current

PROFESSIONAL EXPERIENCE

Full-Stack Web Developer

Whitehall AI Reply, Rome

January 2024 – December 2024

- Developed back-end services in Java / Spring Boot using JAX-RS, REST and SOAP APIs, Oracle DB and JPA/Hibernate for central government clients, following MicroProfile and Agile delivery
- Implemented React front-ends consuming those APIs, coordinated releases via Azure DevOps (Git repos, artifacts, backlogs, tasks) and contributed to feature scoping with client stakeholders

OTHER SKILLS

Python, PyTorch, Pandas, NumPy, SciKit Learn, Transformers, SVM, AdaBoost, Decision Trees, Clustering, CNN, RNN, SQL, Git, Docker.